

Research paper

University students' security behavior against email phishing attacks: insights from the health belief model

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Abstract

Despite advancements in technical defenses, phishing attacks persist due to end-user vulnerabilities and behaviors, including shadow security behaviors like bypassing established protocols. Traditional cybersecurity awareness programs often fail to change behavior effectively because they adopt generic approaches, overlooking the psychological and behavioral factors influencing security practices. To address this, the emerging field of behavioral cybersecurity aims to understand and influence human behavior in cybersecurity contexts. In particular, the Health Belief Model (HBM), traditionally applied in health contexts, has shown promise in predicting cybersecurity behaviors. This study applied an adapted HBM to examine factors influencing university students' security behaviors in response to email phishing attacks. Self-reported data were collected from 569 university students at Universitas Atma Jaya Yogyakarta, Indonesia, using convenience sampling. Confirmatory factor analysis and covariance-based structural equation modeling were employed for data analysis. The results revealed that perceived severity, perceived importance, self-efficacy, and cues to action were statistically significant predictors of security behavior. Students who perceived phishing attacks as severe, recognized the importance of security measures, responded to cues to action, and felt confident in their ability to protect themselves were more likely to engage in security behaviors. However, perceived susceptibility to phishing attacks and perceived barriers to adopting security measures did not show significant relationships with security behavior. However, due to the inherent limitations of self-reported data, the terms "security behavior" and "student security behavior" should be understood as reflective of perceived or intended actions, rather than actual observed behaviors. Despite these limitations, this study provides valuable insights into human-related cybersecurity vulnerabilities by identifying key psychological and behavioral factors influencing security behavior. In doing so, it responds to calls for studies that apply psychological frameworks and models to better understand how individuals respond to security threats and develop more effective protection strategies. Thus, the findings remain relevant and will support the creation of tailored security awareness programs designed to meet the specific needs and characteristics of the studied demographics, ultimately contributing meaningfully to the enhancement of cybersecurity resilience.

Keywords: cybersecurity; email-based phishing attacks; university students; behavioral cybersecurity; health belief Model (HBM)

Introduction

In today's technology-dependent world, we face unprecedented cybersecurity threats, posing significant risks to individuals, organiza-

tions, and governments [1,2]. The swift expansion of email, web-based technologies, smart communication devices, and social media, combined with the growing utilization of artificial intelligence (AI),

has granted cybercriminals the ability to devise increasingly intricate deception tactics, thereby heightening the difficulty of detecting security threats [3]. Among these threats, phishing attacks stand out as a serious menace, serving as one of the most prominent tactics in modern cyber-attacks [4,5]. Phishing entails the deceptive tactics of pressuring and manipulating individuals into divulging sensitive information or assets to unauthorized parties by presenting oneself or the message as legitimate [1,3,4].

This deceptive practice commonly occurs through channels like email, webpages, URL links, instant messaging, or forums [6]. Methods might involve hacking websites, acquiring email lists and botnets, and spoofing emails and SMS to mimic reputable sources, all with the intent of deceiving unsuspecting victims into clicking malicious links or downloading malicious content [7]. The most common type of phishing is phishing emails, wherein legitimate email appearances are replicated to persuade users of their authenticity, prompting them to take the desired action [8]. In these phishing email attacks, deceptive messages are sent to users, with the intent of extracting sensitive data like passwords or credit card information or executing malicious code by opening attachments or visiting websites, thereby granting remote access to the attacker [5]. In general, the goal of any phishing attack is to obtain access to sensitive information, leading to a range of damages for end-users, such as financial losses and damage to their reputation [3,4].

Due to the devastating consequences of phishing attacks, significant technical advances have been made in blocking them [4]. Hardware manufacturers and application programmers strive to embed security features into their products [6]. Currently, widespread technical defenses rely on spam filters and domain blacklisting [5]. However, purely technical defense solutions are deemed insufficient because, despite efforts, millions of scam emails are sent daily, with some managing to evade detection [9]. Detecting phishing attacks automatically is unreliable since they exploit users' vulnerabilities and succeed when users are susceptible to manipulation [3,6]. Thus, users become the last line of defense against these attacks [4,10].

Unfortunately, recent analyses continue to stress that phishing attacks remain widespread and easily executed, affecting individuals across all demographics and social classes, making it a pervasive issue [1]. In 2023, the Anti-Phishing Working Group (APWG) recorded a staggering one million seventy-seven thousand five hundred one phishing attacks in the fourth quarter alone, marking nearly five million phishing attacks throughout the year, the highest volume on record [11]. Signifying that, despite built-in technical security controls, these attacks continue to succeed due to users' unawareness of their vulnerabilities and the inability of technical controls to prevent unwanted user actions that lead to successful attacks [6].

Cybersecurity, a subset of Information Technology (IT) security, which in turn is a subset of Information Security (IS), depends on human users performing appropriate actions that complement the administrative and technical controls already in place [4], referred to as onion defense or defense in depth. Even if technological measures, guidelines, and restrictions are implemented, end users will find a way to circumvent them [12]. Among other reasons, this is because while end-users are often aware of the security threats they may be confronted with, they also must perform their job duties in a timely manner [12]. To comply with both their duties and security policies as much as possible, shadow security behaviors are used [13]. Shadow security behaviors refer to actions taken by individuals that bypass or undermine established security protocols and practices [13]. Such behaviors often arise when individuals prioritize convenience or efficiency over security measures, thereby creating vulnerabilities that can be exploited by attackers. Regarding phishing attacks, shadow

security behaviors can significantly increase the risk of successful attacks. For example, suppose an organization has implemented a policy instructing users not to click on email links from unknown or suspicious senders and to instead verify the legitimacy of the sender before taking any action. However, due to the pressure of meeting deadlines or the urgency of completing tasks, a user receives an email purportedly from a colleague requesting immediate action on a project. Despite the policy, the user might quickly click on the link provided in the email without confirming its authenticity, unknowingly becoming a victim of a phishing attack. In this scenario, the shadow security behavior of clicking on email links without proper verification creates a vulnerability that can be exploited by malicious actors. In this case with a single successful phishing attack, even the most expensive and sophisticated technical controls put in place can be bypassed if an attacker successfully manipulates the end user to open or click a malicious email attachment or link [6]. This underscores the reality that numerous security threats we encounter today cannot be fully addressed solely through technological means [14], and the conduct of end users plays a crucial role in thwarting cyberattacks linked to human actions [12]. One crucial approach is through awareness initiatives that focus on the end users' behavior concerning the threats they might encounter [12].

However, in practice, organizations have often overlooked the importance of focusing on human behavior, and there hasn't been much systematic support for efforts to adopt awareness initiatives in the past [15]. In contrast to this neglect, some organizations across various sectors, including financial services, government, healthcare, and academia, have tried to pay attention by implementing countermeasures such as training programs, awareness campaigns, and simulated phishing attacks. These efforts aim to familiarize end-users with cyber-attacks including phishing attack strategies, help them recognize suspicious emails, and respond appropriately [16,17]. Regrettably, most of the approaches used have often proven ineffective in many instances [18], as they struggle to effectively influence end-user behavior for some reasons [19]. A significant number of these awareness programs adopt a "one-size-fits-all" approach, utilizing uniform materials and lacking customization based on user characteristics [20,21]. These initiatives often rely on conventional methods such as posters, bookmarks, websites, presentations, notes, and occasionally include video materials [12,22–24]. Unfortunately, these methods tend to be predominantly theoretical, with the primary focus aimed at communicating cybersecurity issues only [19,25]. They lack effectiveness in empowering users to recognize and respond to cyber threats, ultimately falling short in altering employee behavior [26]. This signifies that conventional methods of cybersecurity awareness struggle to effectively engage and influence end users' behavior in adopting recommended security measures, leading to low success rates [24,27,28], which is evidenced by the high occurrence of cyber-attacks associated with user behavior [29]. This implies that although cybersecurity awareness initiatives targeting end-users' behavior are important [12], simply conveying information has a restricted impact on altering users' behavior [30], instead, it's a waste of effort and resources [12].

Effective cybersecurity awareness initiatives necessitate users to achieve three crucial stages: perception (recognizing the importance of cybersecurity), comprehension (having a thorough understanding of its significance and personal relevance), and projection (influencing end-users' behavior and ensuring compliance with security protocols post-awareness) [12,19]. According to behaviorism, a psychological theory influencing behavior change can be achieved through conditioning processes, a learning process in which behaviors become associated with specific stimuli or consequences through repeated

exposure [31]. However, before starting the conditioning processes in behaviorism, the first step typically involves identifying the behavior that is being targeted for modification. Meaning, to effectively prevent human related cyber threats, there is a need to understand peoples' behavior [32]. For example, the effectiveness of phishing attacks hinges on specific human behaviors; therefore, effectively defending against phishing attacks requires a deeper comprehension of how human emotions and thoughts shape behavior, ultimately rendering individuals susceptible to phishing attacks [3]. Therefore, more research on how individuals behave in security threat situations is essential to improving protection [4,10,13], and progress in this area can be accelerated if combined with finding ways to influence expected behaviors [33].

Acknowledging that cybersecurity has transformed into a multifaceted domain that encompasses more than just technical aspects [21], the effective solution is not solely reliant on the strength of a technical solution, but rather on end users' behavior [34]. [35] proposes embracing a human-centered approach in cybersecurity, empowering individuals to make informed decisions, adopt secure behaviors, and become active participants in safeguarding against cyber threats. Consequently, drawing on insights from disciplines such as psychology, sociology, and behavioral economics, researchers and practitioners across computing, security, and the social sciences have increasingly turned their focus toward understanding user behavior in the context of cybersecurity [33], an interdisciplinary field known as behavioral cybersecurity. Behavioral cybersecurity has become a vital area of research [33,36]. At its core, behavioral cybersecurity recognizes that the availability, confidentiality, and integrity of information systems are profoundly shaped by the complex interplay of human behaviors (Lahcen et al., 2020). While conventional cybersecurity measures focus on upholding the secrecy, completeness, and accessibility of digital resources, often termed the "CIA triad" in information security [18], research and measures in behavioral cybersecurity aim to understand and protect information systems by considering user psychological, social, cognitive, and emotional aspects [34]. Thus, behavioral cybersecurity can be defined as the study of the multifaceted interactions between human behaviors and information systems, emphasizing how these behaviors impact the availability, confidentiality, and integrity of those systems. It pertains to human behavior and the underlying factors that drive it, encompassing human characteristics, attitudes, motivations, and conditions that shape cybersecurity behavior [33]. Most importantly, findings from behavioral cybersecurity research inform effective cybersecurity awareness programs by providing insights into the underlying factors that influence human behavior in cybersecurity contexts.

As mentioned earlier, most cybersecurity awareness programs employ a "blanket approach" or "one-size-fits-all" strategy and lack justification based on established theories [12]. Consequently, the materials used in most cases are similar [20], without considering how people's beliefs, attitudes, and motivations influence their online behavior, resulting in less engaging, memorable, and ultimately less successful promotion of secure behaviors. It seems decisions made during the preparation process for the materials appear to lack careful consideration, resulting in the frequent use of ineffective training methods [12]. Some of the training initiatives are based on general theories and information, yet it is suggested that a more explicit use of behavior change theories would be more beneficial [12].

Building on the suggestion to consider behavior change theories more explicitly, existing research on phishing attacks susceptibility has progressively applied psychological models and or constructs to examine how users' cognitive and emotional responses shape their vulnerability to phishing attempts. [37] utilized the Extended

Parallel Process Model (EPPM) to examine how perceived severity, susceptibility, and self-efficacy influence college students' attitudes and behavioral intentions regarding phishing. Similarly, [38] employed Structural Equation Modeling to analyze the relationships between psychological factors such as perceived vulnerability, perceived severity, perceived barriers, and self-efficacy, and phishing risk behavior among healthcare staff. In another study, [39] applied the Protection Motivation Theory (PMT) to explore how self-efficacy and protection motivation influence individuals' susceptibility to phishing attacks, particularly in the context of instant messaging. Although limited, other studies have leveraged the Health Belief Model (HBM) to examine user behavior in relation to phishing attacks. For example, research conducted at a Midwestern public master's-granting university [40] used the HBM to investigate email security behaviors among students, faculty, and staff, highlighting differences in their phishing susceptibility and response patterns.

Similarly, in response to calls for research that employs psychological frameworks or models to better understand how people behave in response to security threats and improve protection [4,10,13], this study builds upon prior research by utilizing a psychological model thus, the HBM to explore the cognitive factors (perceived susceptibility, perceived severity, perceived benefits, perceived barriers, cues to action, and self-efficacy) influencing university students' responses to email phishing attempts. This approach aligns with studies such as by [40], which also utilized the HBM to examine email security behaviors. However, unlike [40] "one-size-fits-all" approach, which investigated the email security behaviors of students, faculty, and staff collectively, this study adopts a more focused methodology by targeting students exclusively. This study's focus on university students is justified by the distinct characteristics, roles, and behavioral tendencies that differentiate students from other demographic groups, such as faculty and staff. Faculty and staff often manage sensitive information and critical databases, making them high-value targets for phishing attacks. A successful phishing attempt on these individuals could compromise institutional systems, disrupt administrative operations, and even result in job termination. Their behavior toward phishing is shaped by factors such as professional cybersecurity awareness and training, adherence to organizational IT policies, and the high stakes associated with their roles. In contrast, students typically do not handle such high-stakes institutional data. However, their behavioral patterns in responding to cybersecurity threats are influenced by unique motivational and contextual factors. Students frequently exhibit overconfidence in their digital skills; a characteristic rooted in their classification as "digital natives" due to lifelong exposure to digital technologies. They also constitute the largest and most frequent users of internet-based technologies, with usage rates of 88% compared to other groups [41]. Paradoxically, this confidence often coexists with a lack of formal cybersecurity awareness and training, leaving them particularly vulnerable to phishing attempts. Phishers exploit this vulnerability by crafting convincing emails that mimic legitimate communications from trusted sources, such as university administrators, banks, or popular online platforms. Students may also be more likely to engage with emails promising freebies, discounts, or exclusive offers, making them prime targets for mass phishing, a method favored by hackers for its speed and simplicity [42]. Beyond simple scams, attackers can use email-based phishing schemes to access students' social media accounts, potentially harvesting sensitive or private information for nefarious purposes, including blackmail. Some attackers employ sophisticated schemes such as sextortion, in which students are deceived into believing their explicit content has been compromised [43]. Students' susceptibility also extends to more targeted phishing methods, such as spear

phishing or whaling. For instance, attackers may exploit students whose parents are affluent or influential by using social engineering tactics to compromise their email accounts. These compromised accounts can then serve as gateways for further attacks, enabling fraud or access to valuable financial and personal information. The combination of high email volume, frequent online activity, and the desire to utilize various online services makes university students particularly susceptible to phishing attacks. Studies reveal alarming vulnerability rates: one survey found that nearly 70% of 1 350 students clicked on phishing links [44], while another reported that 70.6% of college students were vulnerable due to a lack of awareness [45]. Given these unique vulnerabilities, combining students with other demographic groups in a single study as was the case in [40], risks obscuring the behavioral nuances specific to each group. The motivations, experiences, and consequences of phishing attacks differ significantly across demographics, necessitating a more focused approach. Moreover, there is a notable gap in research specifically targeting phishing and cybersecurity behaviors among university students [44]. By focusing exclusively on students, this research aims to provide a clearer and more actionable understanding of their behaviors and vulnerabilities. Using the HBM, this study examines the cognitive and emotional factors that influence students' security behaviors concerning phishing attacks. This approach facilitates the identification of common characteristics or beliefs that make this group more susceptible or resistant to phishing scams.

The study relies on self-reported data rather than direct observations, as self-reports are a valuable method for gaining insights into individuals' internal experiences, behaviors, and perceptions, which are often difficult to assess using alternative methods [46–48]. Researchers frequently rely on self-reports to measure psychological constructs because they provide access to internalized information that only individuals themselves can accurately convey [47]. However, self-reports are not without limitations, particularly in terms of data quality and participant compliance [48]. Some of the commonly reported issues will be addressed in the methodology section and further discussed under research limitations.

Despite these challenges, the findings of this study hold significant potential for informing the development of targeted intervention programs designed to improve phishing awareness and enhance security behaviors. Such interventions can strengthen defenses against email-based phishing attacks, promote better overall cybersecurity practices, and drive meaningful behavioral changes among students. Notably, email-based phishing is prioritized due to the central role of email as a communication tool in higher education [49]. Evidence indicates that students conduct a significant portion of course-related communications via email [50], which also happens to be the most common channel for phishing attacks. Approximately 65% of all phishing attempts involve hyperlinks embedded in emails [45]. Addressing this vulnerability is therefore critical to safeguarding students and fostering a secure digital environment within higher education institutions.

Theoretical framework

Health belief model

Introduced in 1952, the Health Belief Model is a psychological framework utilized to understand and forecast health-related behaviors [51], and later applied to understand why individuals either implement or neglect preventive practices [52]. The HBM suggests that a person's perceptions about a health condition influence their response to it [53,54]. This model centers on two key aspects: how

someone perceives the threat of becoming ill and how they assess the actions they can take to safeguard themselves. Initially, the perception of illness threat comprises two beliefs: perceived susceptibility, which refers to one's perceived likelihood of becoming ill, and perceived severity, which pertains to the perceived seriousness of the illness if contracted [53,54]. Secondly, when assessing their behavior, individuals consider the perceived benefits of preventive actions and the perceived obstacles that may impede them [53,54]. Additionally, the model suggests that encountering cues to action, such as reminders or prompts related to a health threat, prompts individuals to take preventive or protective measures in response [53,54]. These measures may include seeking further information, adopting healthy behaviors, scheduling medical appointments, or adhering to prescribed treatments or preventive measures. Put plainly, the HBM proposes that individuals are inclined to take preventive measures if they perceive a personal threat, grasp the severity of the consequences, recognize the benefits of preventive actions, perceive minimal obstacles to adopting those actions, and are prompted by cues to act [55,56]. Nonetheless, individuals are most likely to act if they perceive that the benefits of acting outweigh any barriers or threats [57]. Adjusted version of the HBM also include self-efficacy, borrowed from the Social Cognitive Theory [58], to address the challenges of changing ingrained unhealthy habits. Self-efficacy pertains to an individual's belief in their capacity to execute a particular action or behavior [54,58].

Similarities can be observed between protective security behavior, such as regularly inspecting emails for suspicious content and abstaining from clicking on links or downloading attachments from unfamiliar sources, and preventive healthcare behavior, such as adhering to a healthy diet to prevent illnesses [59]. Both preventive healthcare and protective security behavior involve proactive measures aimed at avoiding adverse outcomes, whether it's prolonging health and mitigating disease effects in healthcare or reducing security risks and the impact of incidents in security practices, by adopting preventive and protective behaviors [59]. Drawing parallels between health-related behaviors and cybersecurity practices, a few researchers have explored the applicability of the HBM in predicting responses to cybersecurity threats. For instance, studies have investigated users' information security behavior [59,60], the adoption of computer security [61], factors affecting user compliance behavior [62], factors driving protective and abusive behaviors in information security [63], and user compliance with security regulations, among others. However, although the HBM offers valuable insights into the determinants of behavior, its applicability in cybersecurity remains limited and yields inconsistent findings. For example, a study by [63] on factors driving both protective and abusive behaviors in information security, conducted within two financial institutions in the US and India, discovered that self-efficacy had a notable correlation with behavior, aligning with a study by [59] on users' computer security behavior. However, in contrast, [63] did not observe any significant correlation between perceived benefits and behavior, whereas [59] results revealed a notable correlation between perceived benefits and behavior. Additionally, in the case of [63] study and [59] study, perceived susceptibility was identified to have a notable correlation with behavior [62] targeting respondents at selected public hospitals in Malaysia, while perceived susceptibility was deemed insignificant. Inconsistencies in research findings, despite similar methodologies and contextual factors, could be attributed to differences in sample size and individual differences among participants, including diverse backgrounds. What affects one demographic clearly is not always applicable to others. Relying solely on others' findings risks overlooking these nuances and may result in ineffective cybersecurity interventions. Therefore, conducting independent research targeting specific

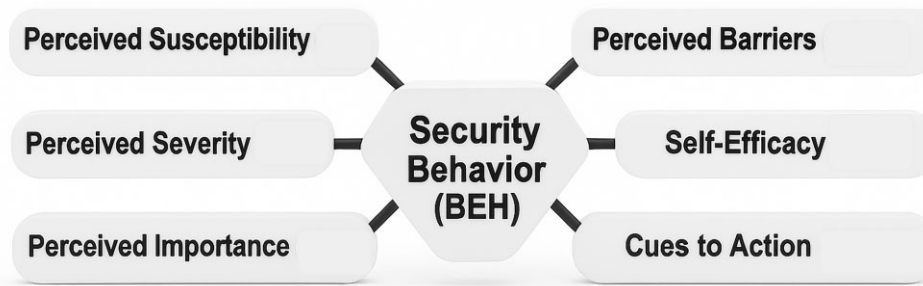


Figure 1. Research Model.

demographics is crucial to uncovering nuanced insights and developing tailored interventions that address the diverse needs of different groups.

Research model and hypothesis development

According to [33], cybersecurity researchers can copy from the health and aviation sectors, which possess extensive knowledge and experience in comprehending human behavior. In line with this recommendation, the study adopts an approach like that of the health industry by utilizing the HBM as previously demonstrated by researchers like [59,62], with necessary adjustments tailored to the specific context under investigation.

Besides the fact that the HBM has been previously used in cybersecurity contexts, this study justifies its use based on several key factors. Firstly, meta-analyses have demonstrated the predictive power of many of its key constructs [64]. Moreover, the HBM incorporates explanatory constructs such as perceived susceptibility, perceived severity, and cues to action, which are absent in other prevalent IT acceptance theories like the Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT), Diffusion of Innovations Theory (DOI), Theory of Reasoned Action (TRA), and Theory of Planned Behavior (TPB), yet are crucial in information security practice [59]. This makes the HBM applicable to explaining a user's information security behavior in an organization [59]. Furthermore, the model proposes that awareness programs designed to foster behavior change are most successful when they target four essential factors: perceived barriers, perceived benefits, self-efficacy. Therefore, in the case of this study, using the HBM to understand which of its factors influence students' behaviors will provide insights that can inform the design of more effective awareness programs. Below, we elaborate on our research model. As depicted in Figure 1, the study utilizes a research model comprising six independent variables: perceived susceptibility (SUS), perceived severity (SEV), perceived importance (IMP), perceived barriers (BAR), cues to action (CUE), and self-efficacy (SEF), along with one dependent variable, security behavior (BEH). The definitions and hypotheses for each variable in the study research model are as follows.

Perceived Susceptibility: In the realm of security, susceptibility is defined as the probability of a security incident occurring [59,61,63]. Individuals tend to engage in preventive behavior more when they perceive a higher possibility of negative outcomes, whereas a lower perceived possibility leads to reduced engagement in preventive measures [63]. However, in the security domain, people may react diversely even when faced with identical situation, which can greatly affect their security behavior, as some might perceive a high

likelihood of an incident while others may deem it improbable [59]. In the context of this study, perceived susceptibility refers to an individual's perception of their vulnerability to email phishing attacks. When individuals perceive susceptibility to a phishing threat, they are more inclined to display increased positive security behavior [59]. Consequently, the study hypothesizes that:

H1

(+): Perceived susceptibility will demonstrate a statistically significant positive correlation with security behavior.

Perceived Severity: In the context of security, perceived severity represents an individual's belief regarding the seriousness of being impacted by a specific security event [63]. It pertains to the perceived level of impact individuals anticipate because of not adhering to recommended cybersecurity policies and procedures. Individuals may perceive consequences ranging from personal repercussions such as blackmail, loss of privacy, job termination to broader organizational impacts like compromised data integrity, system downtime, and company closure [63]. However, despite the severity of potential consequences, individuals may interpret the severity or extent of damage differently [59], leading to varying perceptions of negative outcomes resulting from their actions [63]. Within the scope of this study, we define perceived severity as an individual's perceived consequences of a phishing incident, which should correlate positively with security behavior [59]. Thus, we hypothesize:

H2

(+): Perceived severity will demonstrate a statistically significant positive correlation with security behavior.

Perceived Benefits and Importance: In the HBM, perceived benefits pertain to an individuals' beliefs regarding the relative effectiveness of an action in mitigating the threat of disease [59]. In security domain, it pertains to the belief in the effectiveness of actions required to prevent a security risk [61]. The HBM suggests it will influence their decision to engage in health preventive behaviors [63]. Similarly, in the security domain, if individuals believe in the effectiveness of practicing accepted security practices, it will lead to expected security behavior. However, the importance of perceived benefits lies in their ability to influence behavior. If a person perceives benefits but does not find them important, they are less likely to change. This brings up perceived importance as a relevant construct. Perceived importance refers to the significance that an individual assigns to the outcomes of a specific behavior [65]. The key difference between perceived importance and perceived benefits lies in their focus and evaluation criteria. Perceived importance is about the overall value an individual attaches to the outcomes, which can be positive, resulting from performing the desired behavior, or negative, resulting from not

engaging in behavior, while perceived benefits are specifically concerned with the positive outcomes that result from performing the behavior to mitigate or avoid the threat [66]. Perceived importance refers to individuals' subjective assessment of the significance or importance of the outcomes associated with a behavior. For instance, someone might understand the benefits of multi-factor authentication (MFA) for online security but choose not to enable it due to perceiving it not important. Perceived importance emerges as a better construct, particularly in the security domain where in some cases users perceive security solely as the responsibility of the IT professionals rather than a shared concern [29]. In this scenario, individuals may not recognize the personal relevance or importance of actively participating in security practices themselves. This indicates that even if users understand the benefits of security measures, they may not prioritize them if they do not perceive security as personally important. Therefore, perceived importance, which emphasizes the necessity for individuals to personally prioritize and value security measures, becomes crucial for accurately assessing individual security behaviors, as without this, behavior change may be limited, even among those who recognize the benefits of such practices. Considering this, this research modified the original HBM model by adopting perceived importance as a construct whilst omitting the perceived benefits original HBM construct. The adaptation of the HBM in the security context is not a recent development; previous literature has expanded upon it over the years by introducing additional variables such as general security orientation [59,67] and perceived working experience [62]. Besides, previous research has indicated that perceived importance could also serve as a significant predictor of behavior [67]. In this research, perceived importance is defined as an individual's belief in the importance of taking actions to prevent phishing attacks [67], and the following hypothesis has been formulated:

H3

(+): Perceived importance will demonstrate a statistically significant positive correlation with security behavior.

Perceived Barriers: Perceived barriers refer to obstacles that make it difficult or unpleasant for individuals to perform a specific action, such as following security practices [63]. In the realm of security, adhering to recommended measures often involves additional controls or measures that can be inconvenient or uncomfortable for individuals [59], leading to feelings of discomfort or inconvenience [63]. For instance, complying with security measures may make tasks more challenging or time-consuming, despite their role in safeguarding against attacks [63]. As mentioned earlier, this situation often gives rise to what's known as shadow security behaviors [13], which create vulnerabilities that attackers can exploit. Within the framework of this study, we define perceived barriers as the users' perception of the inconvenience associated with following security best practices [59]. The following hypothesis has been postulated:

H4

(-): Perceived barriers will demonstrate a statistically significant negative correlation with security behavior.

Self-Efficacy: Derived from the social cognitive theory, self-efficacy corresponds to an individual's self-assurance in their capability to execute a behavior [58,63]. In the security context, self-efficacy represents individuals' confidence in their ability to independently perform protective security behaviors. An individual's confidence in taking recommended security precautions, as emphasized by [63], plays a crucial role in mitigating risks, as individuals with high self-efficacy are more inclined to engage in desired behaviors by dedicating more time to comprehending and addressing threats [58]. In

this study, self-efficacy refers to a user's self-assurance in their skills or ability to implement recommended expected security measures, which is expected to enhance security behavior [59]. Therefore, a hypothesis has been framed as follows:

H5

(+): Self-efficacy will demonstrate a statistically significant positive correlation with security behavior.

Cues to Action: In the security context, cues to action comprise external stimuli or prompts that motivate or prompt individuals to participate in recommended security actions or behaviors. Such cues may encompass security campaigns, media prompts, social influences, and recommendations from experts [59,62]. Increased exposure to these cues to action lead to heightened security behaviors [59]. Within the framework of our study, cues to action denote experiences or triggers that motivate and prompt users to engage in expected phishing security behaviors or actions. As a result, the study posits the following hypothesis:

H6

(+): Cues to action will demonstrate a statistically significant positive correlation with security behavior.

Security Behavior: In general, behavior encompasses the actions of organisms within a given environment or context [68,69]. Specifically, in the realm of security, behavior pertains to how individuals act within information systems, encompassing hardware, software, and network systems [70]. Such behavior can manifest as intentionally risky or secure actions [71], and may also occur unintentionally with security implications [70]. Examples of security behavior include password management, handling of organizational data, and utilization of network resources [70]. Depending on the context, users may exhibit desired security behaviors or unwanted, inappropriate behavior [72]. Desirable security behaviors align with recommended practices aimed at thwarting security threats, while unwanted behaviors deviate from these standards [70]. Notably, poor user security behavior often serves as a significant determinant of security incidents [68,73]. In this paper, we define security behavior broadly as the actions and precautions individuals take when using email to safeguard themselves against phishing attacks. Behavior, in its wide scope, involves the conduct of individuals, influenced by numerous factors, both internal and external [68]. This implies that in the domain of cybersecurity, individuals' responses to security threats are shaped by internal factors like perceptions, beliefs, and attitudes, as well as external factors such as inconveniences, awareness campaigns and the effectiveness of security measures to mention but a few. Against this backdrop, the study posits security behavior as a dependent variable, intricately influenced by various factors. Notably, perceived susceptibility, perceived severity, perceived importance, cues to action, and self-efficacy are suggested to have a statistically significant positive effect on security behavior. Conversely, perceived barriers are expected to result in a statistically significant negative impact.

Research methodology

The research adopted a positivist approach, emphasizing empirical evidence to reveal truths about reality [74]. Accordingly, a quantitative methodology was employed, utilizing a close-ended questionnaire based on prior studies related to security behavior using the HBM. The questionnaire functioned as the instrument for gathering data to examine the hypotheses outlined in the proposed model. In the subsequent subsection, we will discuss the research instrument

used, the data collection procedure and study population, as well as the tools and methods employed for data analysis.

Research instrument

The Student Cybersecurity Awareness Survey (SCAS) was devised based on constructs from the Health Belief Model (HBM) [75]. To construct SCAS, a systematic literature review was conducted on studies related to cognitive and emotional factors influencing IT security behaviors, although these were not specifically focused on email phishing attacks. From this review, key constructs from HBM were identified, and relevant questions/items were borrowed directly from existing survey instruments pertinent to the study context (see Appendix A, Table A1) and adapted to suit the context of email phishing attacks. The development process included a content validity assessment to evaluate the clarity, relevance, and appropriateness of the survey items. To ensure that SCAS and its items were clear and easily understood by the target population, the wording and structure of the questions were reviewed and refined based on feedback from an academic expert in quantitative research methodologies (acknowledged in the study). Given that the study was conducted in Indonesia, where the official language is Bahasa Indonesia and many students may not be proficient in English, consultation with a language translation expert was sought to ensure that the meaning of the original English items was accurately conveyed. The survey was translated from English to Bahasa Indonesia.

Before distributing the full survey, cognitive interviewing was conducted with a small group of target participants to assess the feasibility of the study procedures and refine SCAS. Since the number of participants in a pilot study does not require formal statistical calculations. Since the number of participants in a pilot study does not require formal statistical calculations [76], and various experts suggest a minimum threshold for pilot study sample size, such as [77] recommending 10% of the minimum sample projected for the larger study, and [78,79] suggesting a pilot sample size of 12 participants, 12 participants were used in the pilot study. Given that the minimum target respondents for the study were 120 valid responses, and 10% of 120 is 12, the pilot study sample size adhered to the guidelines by [77].

During the pilot test, each participant completed SCAS individually, with the researcher and an assistant translator present. As participants answered the questions, they were encouraged to raise any issues regarding clarity or meaning of specific items. Questions that required clarification were noted for review. Participants were also asked if they encountered challenges with comprehension or interpretation. This method allowed for the assessment of participants' thought processes, ensuring that SCAS items were interpreted as intended.

The feedback collected during the pilot was used to refine SCAS. Specifically, items that participants found unclear or difficult to understand were reworded, and additional explanations were provided for selected constructs. This process also helped mitigate issues related to self-reported data, where unclear instructions or questions could lead to inconsistent responses and undermine data quality [48]. The pilot also provided an opportunity to assess the reliability and variability of SCAS items, ensuring that the instrument would yield meaningful and consistent data in the main study. Refinements improved the clarity, flow, and overall quality of SCAS, while adjustments were also made to survey length to enhance participant experience.

The final version of SCAS was developed as an online questionnaire using Google Cloud Forms and divided into three sections.

Section A included demographic questions such as gender, level of study, field of study, and occupation. Section B evaluated individual exposure to cybersecurity awareness programs at the university, frequency of email usage, prior knowledge of phishing attacks, and prior experience with phishing attacks. Section C assessed participants' perceptions and beliefs regarding email phishing attacks. All items in Section C were formatted on a five-point Likert scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, 5 = strongly agree, allowing participants to indicate their degree of agreement with the statements. By employing this scale, lukewarm midpoint responses were mitigated [80].

Research participants and data collection

The data collection was conducted at Universitas Atma Jaya Yogyakarta, an accredited private tertiary education institution in Yogyakarta, Indonesia, using the convenient sampling method, which involves targeting participants who are readily available and easily accessible to the researcher, often based on their accessibility or ease of reach [81]. As the researcher is a current enrolled student at the university, it was more convenient to conduct the study there. The respondents exclusively comprised university students, including both undergraduates and graduates who are frequent users of emails.

Using printed QR codes, the researcher conducted in-person survey administration, thus personally visited different faculty classes on campus to administer the SCAS, explaining the research aims to students and asking them to scan the code to access the Google Form questionnaire and participate in the study. All respondents were assured of the confidentiality of their responses, and it was emphasized that their data would be strictly used for academic purposes only, ensuring their privacy and adherence to research ethics. The actual number of students asked to participate in the survey is unknown; however, the research aimed for a minimum sample size of valid responses of 120, determined by a ratio of 20:1, thus 20 valid responses for each construct in the theoretical model [82,83].

Data analysis tools and methods

To analyze the data, the study utilized the Statistical Package for the Social Sciences version 27 (SPSS 27) and Analysis of Moment Structures version 23.0 (AMOS 23.0) version 23.0 graphical software from International Business Machines Corporation (IBM). Initial screening for missing values, unengaged responses, and outliers was conducted using SPSS 27.

SPSS 27 was employed to ensure that basic conditions necessary for certain statistical methods or models to be valid and reliable were met through Cronbach's alpha reliability test, normality test, multicollinearity test, and common method bias assessment. Confirmatory Factor Analysis (CFA) and Structural Equation Modelling (SEM) were carried out using AMOS 23.0, as guided by [84].

CFA was chosen over Exploratory Factor Analysis (EFA) due to the theory-driven nature of the study [85]. EFA is an empirical process used to determine the number of latent constructs and the underlying factor structure of measured variables, facilitating the identification of patterns in the data without testing predefined notions or theories. Conversely, CFA is employed to validate predetermined hypotheses about the latent constructs that govern the relationships among observed variables [85], making it the most relevant approach for this research. Besides CFA has few advantages over EFA. In EFA, factors may or may not be allowed to correlate with each other, while in CFA, the investigator specifies whether individual factors can correlate with each other [85].

SEM also referred to as latent variable analysis or covariance structure analysis, seeks to elucidate the connections between latent constructs indicated by diverse measures [86]. It provides a holistic model that includes the dependence and interdependence among these constructs, revealing the complex relationships within the data. By analyzing covariance statistics, it explains how variables co-vary and vary together, providing insights into their relationships [86]. Two methods of SEM analysis include Covariance-Based Structural Equation Modeling (CB-SEM) and Variance-Based Partial Least Squares Structural Equation Modeling (PLS-SEM) [86,87]. PLS-SEM is geared towards prediction, particularly in exploratory research. PLS-SEM is utilized when researchers seek to explore new relationships or when they have limited prior knowledge about the underlying theoretical constructs [87]. It prioritizes the ability to predict outcomes rather than testing specific hypotheses derived from existing theories [86]. In contrast CB-SEM is often employed to confirm existing theories, aiming to explain relationships between variables [86,87]. It focuses on testing hypotheses derived from established theoretical frameworks [86]. If the aim of the research is theory testing and confirmation, CB-SEM is the suitable method, whereas for prediction and theory development, PLS-SEM is more appropriate [86]. Since the research aimed at confirming established model and testing specific hypotheses, CB-SEM was chosen as the appropriate method for data analysis.

Results

Data cleaning

Data cleaning was conducted to ensure the accuracy and reliability of the analysis by identifying and removing outliers using SPSS 27. Outliers, defined as data points that deviate significantly from the rest of the sample or majority of the data [88,89], can be substantially smaller or larger than other observations in a dataset [88]. If left untreated, outliers pose a risk to the integrity of statistical results, potentially distorting interpretations and leading to erroneous conclusions [88,89]. In statistical analysis, outliers are commonly identified based on Z-scores, which measure a value's relationship to the mean of a dataset by indicating how many standard deviations the value is from the mean [88]. While the acceptable range for Z-scores can vary depending on context and researcher criteria, a Z-score between -2 and 2 is typically considered acceptable. However, more stringent criteria may involve thresholds of -3 to 3 or -4 to 4 standard deviations for identifying outliers [88]. In this study, a threshold of greater than + 3 or less than -3 was applied, resulting in the removal of 34 data points with Z-scores outside this range, leaving 535 data points for further analysis. This step was essential to enhancing the validity of the findings by minimizing the impact of extreme values.

Descriptive analysis

A few important demographic factors, such as gender, study level, and nature of study program, as well as situational variables, including receipt of cybersecurity awareness training, email use frequency, knowledge of phishing attacks, and victimization by phishing attacks, were examined (see Appendix B, Table B1). Firstly, concerning gender distribution, the sample consisted of 244 males (45.6%) and 291 females (54.4%). This gender distribution indicates a relatively balanced representation of both sexes within the sample. Secondly, regarding the study level of the respondents, the vast majority were pursuing bachelor's degrees, comprising 514 individuals (96.1%), while a smaller proportion, constituting 21 respondents (3.9%), were enrolled in master's programs. The study program's nature further

Table 1. Multicollinearity results.

Model		Tolerance	VIF
1	AvSUS	.845	1.184
	AvSEV	.618	1.619
	AvIMP	.551	1.816
	AvBAR	.763	1.311
	AvSEF	.922	1.085
	AvCUE	.562	1.781

a: Dependent Variable: AvBEH.

elucidated the educational backgrounds of the respondents. A significant portion, consisting of 205 individuals (38.3%), were enrolled in IT-related programs, while the majority, encompassing 330 respondents (61.7%), pursued non-IT related programs. This distribution highlights a diverse representation of academic disciplines within the sample. A critical aspect of the analysis pertained to cybersecurity awareness training provided by the university. Approximately half of the respondents, accounting for 267 individuals (49.9%), reported not having received cybersecurity awareness training, while the remaining 268 respondents (50.1%) confirmed receiving such training. This finding underscores the importance of cybersecurity education initiatives within academic institutions. Email usage frequency among respondents varied considerably. The majority reported using email infrequently (42.4%), followed by several times a week (33.8%) and daily (14.6%). A smaller proportion reported using email once a week (8.2%), while only a negligible percentage reported never using email (0.9%). Regarding knowledge of phishing attacks, a significant majority of respondents (86.3%) indicated awareness of such attacks, while a minority (13.6%) reported a lack of knowledge. This suggests a relatively high level of awareness among the sample population regarding this prevalent cybersecurity threat. Finally, concerning victimization by phishing attacks, the majority of respondents (76.8%) reported not being victims of such attacks, while 23.2% indicated experiencing phishing attacks. This finding underscores the existence of phishing attacks within the population and the need for cybersecurity measures to mitigate their impact.

Multicollinearity analysis

Multicollinearity occurs when independent variables in a predictive model overlap, making it tricky to assess each variable's unique impact [90]. Variance Inflation Factors (VIFs) help detect multicollinearity in regression analysis by measuring how much the variance of predictor coefficients is inflated due to correlations between predictors [90]. Higher VIF values signify stronger correlations among predictors, where values below 1 imply no correlation, values ranging from 1 to 5 suggest moderate correlation, and values exceeding 5 indicate high correlation [90]. In this study, VIF values as shown in Table 1 slightly exceed 1 for all predictor variables, indicating that multicollinearity isn't a significant concern among these variables in the regression model.

Common method bias

As highlighted earlier in the introduction, due to this study reliance on self-reported data, issues such as common method bias may arise. Common method bias occurs when research outcomes are flawed as a result of the measurement method used, rather than the constructs under investigation [91]. To verify that the instrument did not negatively affect the quality of the data, a *post hoc* method, Harman's

Single Factor Test, was conducted on the collected data to check whether method bias might have influenced the results. Harman's Single Factor Test assesses if a single factor accounts for over 50% of the variance, suggesting the presence of potential common method bias [91–93]. A result below 50% indicates the absence of significant common method bias [91–93]. The study test revealed a variance proportion of 27.650%, indicating a lesser degree of common method bias and that multiple factors are needed to explain the variance, suggesting the absence of common method biases [93]. This helps ensure accurate data measurement and analysis.

Confirmatory factory analysis

An initial CFA model was carried out to assess the factor loadings (standard regression weights). According to [94], factor loading estimates should ideally surpass 0.5, with values of 0.7 or higher considered optimal, implying that the observed variables have a stronger association with the construct they are designed to assess. Additionally, [95] suggests that items with factor loadings below 0.5 should be discarded, as weaker associations may not effectively represent the underlying constructs. Table 2 illustrates the CFA results, showing that the factor loadings of three items, SUS4, BAR1, and BAR2, are below the 0.5 threshold; as a result, they were removed. To evaluate convergent validity (CV), it's essential to compute factor loadings, composite reliability (CR), and average variance extracted (AVE) [87]. Hence, the obtained factor loadings from CFA were utilized to assess convergent validity, and discriminant validity after removing factors with loadings below the 0.5 threshold to ensure that only the strongest contributing items are considered, providing a clearer understanding of the relationships between items and constructs.

Data reliability, convergent and discriminant validity

Data reliability concerns the consistency and stability of measurements or responses obtained from a dataset, indicating how accurately the data reflect the underlying construct being measured [96]. In this study, we used Cronbach's alpha method to assess data reliability. This statistical measure evaluates the internal coherence or reliability of a test or scale, showing how well the items within it are correlated with each other in measuring the same underlying construct [96,97]. Cronbach's alpha values vary between zero and one, with higher values signifying greater internal consistency among the items [96,98]. Different ranges of alpha values are associated with descriptors like excellent, strong, reliable, moderate, and low, providing insights into the degree of consistency and reliability of the test or scale in measuring the intended construct [97]. Cronbach alpha was calculated after discarding items with loading below threshold and the outcomes for each construct, as displayed in Table 3, exceeded 0.7, indicating excellent reliability based on the qualitative descriptors by [99], where values above 0.9 represent excellent, 0.8 to less than 0.9 represent very good, 0.7 to less than 0.8 represent good, 0.6 to less than 0.7 represent moderate, and less than 0.6 represent poor reliability. Convergent validity evaluates the correlation among items meant to measure the same construct, utilizing factor loadings from CFA to calculate AVE and CR [100]. AVE reflects the proportion of variance explained by a latent variable relative to variance stemming from measurement error. Essentially, AVE gauges how closely the items or indicators of a construct correlate with each other compared to the variance attributed to measurement error, with a threshold of 0.5 or greater indicating convergent validity [100,101]. Higher AVE values suggest that most variation in questions pertains to the

Table 2. Confirmatory factor analysis results.

Construct	Identifier	Item	Factor loading	Verdict
Perceived susceptibility	SUS	SUS1	0.517	Accepted
		SUS2	0.787	Accepted
		SUS3	0.760	Accepted
		SUS4	0.482	Removed
		SUS5	0.558	Accepted
Perceived severity	SEV	SEV1	0.679	Accepted
		SEV2	0.769	Accepted
		SEV3	0.690	Accepted
		SEV4	0.584	Accepted
		SEV5	0.655	Accepted
Perceived barriers	BAR	BAR1	0.333	Removed
		BAR2	0.409	Removed
		BAR3	0.751	Accepted
		BAR4	0.840	Accepted
		BAR5	0.722	Accepted
Self-efficacy	SEF	SEF1	0.823	Accepted
		SEF2	0.797	Accepted
		SEF3	0.690	Accepted
		SEF4	0.652	Accepted
Cues to action	CUE	CUE1	0.589	Accepted
		CUE2	0.755	Accepted
		CUE3	0.756	Accepted
		CUE4	0.783	Accepted
		CUE5	0.796	Accepted
Perceived importance	IMP	IMP1	0.812	Accepted
		IMP2	0.817	Accepted
		IMP3	0.863	Accepted
		IMP4	0.679	Accepted
		IMP5	0.850	Accepted
Security behavior	BEH	BEH1	0.831	Accepted
		BEH2	0.879	Accepted
		BEH3	0.722	Accepted
		BEH4	0.764	Accepted
		BEH5	0.611	Accepted

construct of interest, consistent with the recommendation of [101]. On the contrary, although AVE serves as a cautious measure of the alignment between survey items and the underlying concept, it might exhibit excessive stringency. Thus, solely evaluating CR could suffice in determining the reliability [101,102].

Consequently, if the CR score meets the criteria, it signifies that the survey adequately captures the construct, even in instances where the Average Variance Extracted score falls short of acceptability. Additionally, if AVE falls below 0.5 but CR exceeds 0.6, CV remains acceptable [101,103]. CR assesses the consistency of survey items measuring the same construct, with acceptable values ranging from 0.6 to 0.7 [100,101]. Higher CR values indicate greater reliability and consistency in measurement, suggesting that the construct explains a significant portion of the observed variable's variability [100,102]. Finally, to assess whether different constructs measure distinct concepts, discriminant validity is assessed by contrasting the square root of AVE with inter-construct correlations, validating it if the square root of AVE exceeds correlations between constructs. The findings presented in Table 3 confirm convergent validity through the exam-

Table 3. Reliability, convergent & discriminant validity results.

Construct	α	N of Items	CR	AVE	SUS	SEV	BAR	SEF	CUE	IMP	BEH
SUS	0.751	4	0.904	0.445	0.667						
SEV	0.804	5	0.956	0.460	.167**	0.678					
BAR	0.751	3	0.925	0.596	.379**	.256**	0.773				
SEF	0.828	4	0.965	0.553	-0.037	.009	-0.178**	0.744			
CUE	0.853	5	0.954	0.550	.207**	.520**	.295**	.109 ^a	0.74		
IMP	0.899	5	0.979	0.651	.105 ^a	.556**	.165**	.156**	.594**	0.807	
BEH	0.865	5	0.980	0.585	.105 ^a	.503**	.127**	.225**	.538**	.688**	0.765

Notes: Cronbach's alpha (α); Number (N) after discarding items with loading below threshold;

**means Correlation is significant at the 0.01 level (2-tailed);

^aCorrelation is significant at the 0.05 level (2-tailed).

Table 4. Proposed hypothesis results from SEM analysis.

Proposed hypothesis	Effect	Total effect	Verdict
H1: Perceived susceptibility will demonstrate a statistically significant correlation with security behavior	Direct	-0.018NS(-0.017S)	Rejected
H2: Perceived severity will demonstrate a statistically significant correlation with security behavior.	Direct	0.109*(0.108M)	Accepted
H3: Perceived importance will demonstrate a statistically significant correlation with security behavior	Direct	0.598***(-0.600L)	Accepted
H4: Perceived barriers will demonstrate a statistically significant negative correlation with security behavior	Direct	-0.026NS(-0.028S)	Rejected
H5: Self-efficacy will demonstrate a statistically significant positive correlation with security behavior	Direct	0.209**(-0.153M)	Accepted
H6: Cues to action will demonstrate a statistically significant positive correlation with security behavior.	Direct	0.083**(-0.107M)	Accepted

ination of AVE, CR, and the square root of AVE, while also indicating distinctiveness among constructs, thereby supporting the validity of the research model. Nevertheless, it is noteworthy that two constructs, SUS and SEV, exhibited AVE values below the 0.5 threshold. However, both constructs exhibited CR values surpassing 0.9, significantly exceeding the recommended threshold, indicating satisfactory internal reliability of the measurement items [101,102].

Normality analysis

Normality analyses determine whether a dataset follows a normal distribution, which is crucial for choosing the right statistical methods in research [104]. Skewness and kurtosis tests are commonly used to assess normality, with skewness indicating data symmetry and kurtosis measuring distribution peakedness compared to a normal distribution [104,105]. Data considered normal usually have skewness values ranging from -2 to +2 and kurtosis values ranging from -7 to +7 [94], while values exceeding 3 for skewness and 10 for kurtosis may indicate issues with the data [82,83]. If the data are normally distributed, parametric analyses may be suitable; otherwise, alternative methods may be needed for accurate analysis [104]. SEM relies on assumptions like sufficient sample size and normal data distribution for parametric analyses [106]. According to [107], skewness values between -3 and +3 and kurtosis values between -10 and +10 are deemed suitable for SEM analysis. The study skewness values ranged from -1.027 to 0.130, and kurtosis ranged from -0.727 to 0.992, falling within acceptable ranges for SEM analysis. This indicates that the data meet the necessary conditions for SEM analysis.

Proposed research model analysis

To reiterate, the proposed model analysis utilized SEM, a quantitative method for exploring relationships among different

factors [82,83]. SEM offers flexibility in analyzing data and theories, allowing researchers to investigate connections between predictor and criterion variables, create latent variables, account for measurement errors, and assess theoretical assumptions using empirical data [95].

Thus, SEM was employed to test the hypotheses proposed in the theoretical model and evaluate how well the model represents the relationships among variables based on collected data, determining its fit. The SEM analysis results, depicted in Table 4, illustrate the direct effects of each independent variable on the dependent variable (BEH). These effects are presented in a specific format: first, the unstandardized effect (β), followed by statistical significance (p) denoted by *, **, ***, and NS to signify significance levels of 0.05, 0.01, 0.001, and not significant, respectively. This is followed by the standardized effect and its magnitude provided in parentheses, with the magnitude categorized as small (S) for ≤ 0.1 , medium (M) for between 0.1 and 0.5, or large (L) for ≥ 0.5 [108]. Overall, the SEM results in Table 4 indicate that all direct effects in the research model are statistically significant, except for perceived susceptibility and perceived barriers.

In addition, as per various prior model fit interpretations, the model fit statistics from the research model in Table 5 appear satisfactory based on several criteria. The Normed Chi-square (χ^2/df) falls within the reasonable range of 1 to 5, indicating acceptable fit. Additionally, the Root Mean Square Residual (RMR) values, approaching 0, suggest a good model fit. Moreover, the values of the Goodness-of-Fit Index (GFI), Normed Fit Index (NFI), Incremental Fit Index (IFI), and Comparative Fit Index (CFI) nearing 1 indicate a close fit, with values above 0.80 implying a good fit. Although the Root Mean Square Error of Approximation (RMSEA) slightly exceeds the conventional cutoff of 0.05 with a value of 0.06, scholars such as [109] and [110] propose 0.06 as an acceptable threshold for indicating a

Table 5. Proposed model fit statistics.

Model Fit Criteria	Model statistics	Interpretations	References
Sample size	535	n/a	n/a
Explained variance of BEH (R-squared)	0.645	n/a	n/a
Normed chi-square (NC) = χ^2/df	2.991	Good model fit	[83]
Root mean square residual (RMR)	0.039	Good model fit	[83,94]
Goodness of fit index	0.870	Acceptable model fit	[131,132]
Normed fit index	0.862	Acceptable model fit	[132]
Incremental fit index	0.904	Good model fit	[83]
Comparative fit index	0.903	Good model fit	[83]
Root mean square error of approximation	0.060	Good model fit	[109,110]

good fit, which appears to be widely agreed upon by experts in this field, further supporting the adequacy of the model fit.

Discussions

In this section, the study discusses the empirical findings, offering possible explanations for why they either support or contradict the initial hypotheses, while also comparing them with prior literature in which the HBM model was used to predict security behavior in IS for additional context and insights. To ensure a clear and organized discussion, the discussions are systematically structured, starting from the first hypothesis and progressing sequentially through to the last, as below.

H1:

Perceived susceptibility will demonstrate a statistically significant positive correlation with security behavior.

The findings indicate a lack of statistically significant relationship between perceived susceptibility to phishing attacks and expected security behavior among university students (H1; $\beta = -0.018$; $p = \text{NS}$). In other words, perceived susceptibility to phishing attacks does not significantly influence how the university students behave in terms of security measures against email phishing attacks. This finding is surprising for two main reasons: 1) The HBM emphasizes this construct as a significant predictor of behavior. 2) Previous research in information security employing the HBM model has consistently highlighted perceived susceptibility as a significant direct predictor of security behavior across diverse contexts, such as users' computer security behavior in exercising care with email attachments [59], employees' information security behaviors [60,63], and end-user motivation to utilize computer security software [61]. However, our research results align with studies by [62], where perceived susceptibility was found to be insignificant in predicting user behaviors towards security policy compliance, and the likelihood of performing phishing attack prevention behavior [67].

Several reasons may explain the non-significance of perceived susceptibility in our research findings. Firstly, the majority of participants reported using emails less frequently. With lower email exposure, individuals may perceive themselves as less susceptible to phishing attacks, potentially influencing the variable's significance in the analysis. Additionally, limited comprehension of phishing attacks among university students due to the lack of cybersecurity awareness

programs may lead students to underestimate their vulnerability, as indicated by the descriptive analysis results, where nearly half of the respondents reported not having received such awareness from the university. Moreover, according to the descriptive analysis, less than a quarter of the students were victims of phishing attacks. It can be argued that students who lack previous experience with phishing attacks may not recognize the extent of their vulnerability, further impacting the significance of perceived susceptibility in the analysis. Although psychological effects of cyber-attacks are often overlooked in research and practice [111], this can be supported by trauma theory. Trauma theory posits that events perceived as harmful, threatening, or overwhelming by the individual experiencing them can have lasting effects on their functioning, including disruptions to their sense of safety, stability, and worldview, both physically, emotionally, and psychologically [112,113]. While phishing attacks may not always cause physical harm and victims face practical issues such as financial loss, identity theft, and damage to reputation, they can have significant psychological and emotional consequences. The emotional and psychological consequences ensuing from the losses associated with such attacks may vary from mild to severe, inducing feelings of fear, anxiety, vulnerability, betrayal, distrust, depression, post-traumatic stress disorders, and panic attacks [114], and even suicide as portrayed in a fictional but plausible movie titled "The Beekeeper, 2024," where a kind-hearted landlady dies by suicide after losing her savings to a phishing scam [115]. Although PTSD is widely recognized as a mental health medical disorder, it also represents understandable responses to traumatic events [116]. Thus, in some cases, phishing attacks can lead to cyber-related post-traumatic disorders (Cyber PTSD) [117], a condition that develops after experiencing or witnessing a traumatic event and is more likely when security breaches involve personal data [118]. Individuals with Cyber PTSD may develop heightened perceptions of vulnerability to similar events in the future, exacerbating feelings of fear, anxiety, and hypervigilance. In addition, experiential learning theory posits that individuals acquire knowledge, skills, and attitudes through direct experience and reflection on that experience [119]. According to this theory, personal experiences play a crucial role in shaping individuals' awareness and understanding of various phenomena [119]. Prior experiences of phishing attacks can contribute to increased perceived susceptibility, as argued by [61], who suggests that such experiences significantly influence individuals' perceptions and persuade them to take precautions. In contrary, without direct encounters with phishing incidents, individuals might underestimate the likelihood of falling victim to these attacks, leading to reduced perceived susceptibility. Consequently, this lowered perception of vulnerability may diminish the significance of perceived susceptibility as a predictor of behavior.

Moreover, optimism bias, which involves individuals' tendency to overestimate the probability of positive outcomes and underestimate the probability of negative outcomes, could also elucidate this finding [120]. In essence, individuals frequently perceive themselves as less vulnerable to negative events compared to others, while concurrently considering themselves more prone to experiencing positive events rather than negative ones [120]. This bias can influence decision-making, risk perception, and behavior in various domains including the information security. Moreover, optimism bias can reinforce the prevailing belief that phishing attacks primarily target affluent individuals or those closely associated with high-ranking figures in multi-million-dollar companies, known as spear phishing. This might contribute to a sense of unrealistic optimism among university students, particularly in a developing country like Indonesia. This perception, coupled with the observation that most reported phishing incidents occur in developed countries with higher per capita income, can

foster false sense of security among students, especially in a student-centric city like Yogyakarta with relatively lower per capita income. Consequently, students may harbor the misconception that they are immune to phishing attacks due to surviving on a low allowance from their parents, and that negative consequences are unlikely to affect them, supported by the HBM theory. Besides, in the fourth quarter of 2023, Indonesia received 8 161 phishing reports, marking a decrease of 1 662 reports compared to the third quarter of the same year [121]. This decline in reported incidents might inadvertently foster a sense of complacency among individuals, leading to a lowered perception of susceptibility to phishing attacks. This decrease in reported incidents could lead individuals to perceive that phishing attacks are becoming less prevalent or less of a threat. Subjective norms could further contribute to this perception. If individuals observe that others around them are not as concerned about phishing attacks due to the decrease in reported incidents, they may also adopt a similar attitude of reduced concern or vigilance towards phishing threats. This alignment with perceived social norms could reinforce the sense of complacency and lower perceived susceptibility to phishing attacks among students.

The discrepancy in perception between students in IT-related programs and those in non-IT related programs could account for the non-significance of perceived susceptibility finding. Descriptive analysis shows that the majority of students are enrolled in non-IT related programs, potentially leading to less exposure to cybersecurity concepts and an underestimation of vulnerability to attacks compared to their IT-related counterparts. This observation aligns with Social Cognitive Theory, which suggests that individuals' perceptions and behaviors are influenced by their environment. In this case, the disparity in perception between students enrolled in IT-related programs and those in non-IT related programs reflects the influence of their educational environment. Students in IT-related programs are likely exposed to more cybersecurity concepts and discussions, fostering a deeper understanding of security risks. Conversely, students in non-IT related programs may have limited exposure to these concepts, resulting in an underestimated susceptibility to attacks. This disparity also underscores how environmental factors, like educational background, shape individuals' perceptions and behaviors regarding cybersecurity. Conversely, students in IT-related programs may possess a deeper understanding of cybersecurity risks, perceiving themselves as more susceptible to phishing attacks. This contrast in awareness between the two groups might underlie the non-significance of perceived susceptibility, as program type influences students' awareness and understanding of security risks, thereby shaping individual perceptions of susceptibility.

However, despite our proposed explanations for the absence of a statistically significant positive correlation between perceived susceptibility and security behavior, it's notable that our results indicate an indirect (negative) non-significant relationship between the two. This suggests that as perceived susceptibility to phishing attacks rises (or falls), security behavior against email phishing attacks declines (or improves), respectively. Nevertheless, since the coefficient lacks statistical significance, this relationship may not apply in practical scenarios.

H2:

Perceived severity will demonstrate a statistically significant positive correlation with security behavior.

The result showed that perceived severity had a significant positive effect on security behavior (H2; $\beta = 0.109$; $P = 0.05$). This suggests that individuals who perceive a phishing attack and its consequences as highly serious are more inclined to engage in phishing

attack prevention behavior [62,67]. One possible explanation for severity being significant while susceptibility is not could be that students are more acutely aware of the potential consequences of phishing attacks than they are of the likelihood of being targeted by such attacks. Specifically, students are likely aware that phishing attacks can compromise their privacy, including accessing sensitive or personal conversations stored in their email accounts and social media accounts to their email, leading to embarrassment, reputational damage, or even personal harm if the content is shared without consent. Since students may perceive the severity of these consequences to be high, they may be more motivated to take preventive measures, resulting in a significant relationship between severity and behavior. This emphasizes the importance of understanding the potential consequences of security attacks, as highlighted in previous research [62]. Additionally, while the significance of perceived severity is not surprising based on previous studies in information security, it contrasts with some surprising findings where it was non-significant in relation to security software usage [61] and security behavior intention [63].

H3:

Perceived importance will demonstrate a statistically significant positive correlation with security behavior

The results support the hypothesis (H3; $\beta = 0.598$; $P = 0.001$). Confirming that individuals who recognize the significance of taking proactive measures are more inclined to engage in behaviors aimed at preventing phishing attacks [67]. In other words, if students understand the importance of safeguarding themselves against phishing threats, they are more likely to take steps to protect themselves from falling victim to such attacks. It's plausible that the recognition of importance could be influenced by the perceived severity of phishing attacks. Considering that students are aware of the potential consequences, including compromising their privacy by accessing sensitive or personal conversations stored in their email accounts, it's reasonable to assume that they grasp the severity of potential outcomes like embarrassment, reputational damage, or personal harm if their private content is shared without consent. In light of this understanding, students may be inclined to perceive proactive behavior as crucial to mitigate the severity of these potential consequences. It's worth noting that the research finding aligns with prior research by [59,67].

H4:

Perceived barriers will demonstrate a statistically significant negative correlation with security behavior.

The study did not find evidence to support the idea that perceived barriers lead to lower security behavior (H4; $\beta = -0.026$; $p = \text{NS}$), resulting in the rejection of the hypothesis. Security is usually viewed as an inconvenience, which may deter users from practicing safe behavior [59]. In contrast, the finding suggests that perceived barriers, encompassing factors like inconvenience, obstacles, or challenges in adopting security measures, do not exert a significant impact on students' behavior in dealing with phishing attacks. The descriptive analysis reveals that while the majority of students (61.7%) are enrolled in non-IT related programs, there is still a presence of IT-related programs (38.3%) students. Therefore, the presence of students studying IT-related programs and potentially IT-savvy individuals within the non-IT programs may explain this finding, as they may not encounter challenges in implementing anticipated security measures [59]. Consequently, the presence of IT-savvy students across study programs could indirectly influence the significance of perceived barriers in shaping security behavior regarding phishing attacks. Further exploration is warranted to understand how students' program backgrounds interact with their perceptions of barriers and

to assess how one's IT savvy (Digital Literacy) affects behavior regardless of study program threats such as phishing attacks.

Moreover, as will be elaborated shortly, the findings also underscore the significance of self-efficacy or self-assurance ($\beta = -0.209$, $P = 0.01$). Self-assurance in ability to identify and report phishing emails enhances the likelihood of proactive protection against such attacks [122]. Additionally, this not only influences individuals' likelihood of acting but also helps them overcome perceived barriers to practicing good security behavior [123]. This may also explain the non-significance of perceived barriers, as individuals who are confident in their ability to recognize and report phishing emails may perceive fewer obstacles to practicing good security behavior. Their confidence in their skills and knowledge may outweigh any perceived barriers, resulting in a non-significant relationship between perceived barriers and security behavior. Furthermore, individuals with high self-efficacy may actively seek solutions to overcome any perceived barriers they encounter, further diminishing their impact on behavior. Therefore, the presence of high self-efficacy can mitigate the influence of perceived barriers, resulting in their non-significance in predicting security behavior. Interestingly, our findings diverge from previous research that supported our proposed hypothesis, indicating a demotivating effect of barriers on phishing attack prevention behavior [67] and exerts a notable adverse impact on the utilization of computer security [61]. Similarly, barriers such as inconvenience and time limitations have been shown to negatively impact employees' intention to practice security behaviors [60] and security and policy compliance behavior [62,63]. Despite these disparities, our findings align with prior studies in affirming the negative effect ($\beta = -0.026$) of perceived barriers on security behavior, although it is non-significant ($p = NS$).

H5:

Self-efficacy will demonstrate a statistically significant positive correlation with security behavior.

The results indicate that self-efficacy plays a statistically significant role in shaping users' security behavior ($\beta = -0.209$, $P = 0.01$) and supports the proposed hypothesis. This implies that individuals' belief in their ability to protect themselves against phishing attacks significantly influences their security behavior regarding email phishing attacks [122]. Given that roughly half of the respondents received cybersecurity awareness from the university, it's plausible to speculate that this exposure to cybersecurity education could contribute to the significance of self-efficacy. Individuals who have undergone cybersecurity awareness training may experience increased confidence in their capacity to identify and address phishing threats, thereby influencing a positive security behavior. Generally, the finding is not surprising, as previous studies have affirmed the significant influence one's confidence on the likelihood of engaging in preventive behavior [59,61–63,67]. Hence, boosting user confidence through consistent cybersecurity training and skill development is vital to prompt users to adopt the suggested measures.

H6:

Cues to action will demonstrate a statistically significant positive correlation with security behavior.

The results, supported by previous studies [60,62,123], confirm the hypothesis (H3; $\beta = 0.083$; $P = 0.01$), indicating that individuals are more likely to engage in security behaviors when they receive prompts or signals encouraging action, such as reminders, alerts, or warnings about potential security threats [60]. These cues act as triggers prompting individuals to adopt protective measures against security threats, thereby enhancing overall security

behavior. The data from descriptive analysis indicates that roughly half of the respondents received cybersecurity awareness from their university, suggesting that a significant portion of the sample had exposure to information about cybersecurity threats and protective measures. Consequently, those who received cybersecurity awareness may be more receptive to cues or prompts related to security threats, making them more likely to respond to such cues and engage in security behaviors. Conversely, for those who have not received cybersecurity awareness, their response to cues may be influenced by social factors rather than their own understanding of the risks. They may observe others taking security measures and follow suit to fit in or please those around them, rather than fully comprehending the importance of the action. This highlights the role of subjective norms or social influence, where the desire for acceptance or approval from peers or authority figures significantly shapes behavior in the absence of formal cybersecurity awareness [123].

Limitations and future work

This study has several limitations. Although the terms “security behavior” and “student security behavior” have been used throughout this paper, it is important to note that these reflect participants' self-reported intentions, attitudes, or perceptions about security behavior, rather than direct observation of actual behavior, which introduces inherent limitations. While self-report data provides valuable insights into individual experiences and perceptions, it is crucial to acknowledge the potential discrepancies between stated behavioral and actual behavior [124,125]. One key factor contributing to such discrepancies is social desirability bias, where respondents tend to answer questions in a way that is perceived as more socially acceptable or favorable, rather than providing a truthful representation of their actual beliefs, attitudes, or behaviors [126,127]. In a similar vein, though arising from different motivations, self-reported data is also subject to judgment of plausibility, where individuals offer explanations for their actions or behavior based on what they perceive to be logical or sensible, rather than recalling the actual reasons behind their behavior [128]. Additionally, numerous studies based on self-reported data have highlighted a tendency for individuals to provide overly optimistic assessments of their own actions, abilities and characteristics, a phenomenon known as biased self-assessment [129]. For instance, a review of studies on how accurately doctors assessed their own skills revealed that only a small number showed a moderate positive correlation between self-reported assessments and actual performance, while most studies found little to no connection or even negative correlations [129]. In addition to the biases inherent in self-reports, various factors related to the data collection instrument can also affect the accuracy of self-reported data. These factors include the wording and format of the questions, as well as contextual cues [130], which may lead respondents to interpret survey questions differently. This can result in biased, inconsistent, or inaccurate response [125,130]. Even with a well-designed instrument, participants may not carefully read or fully understand the instructions or questions. Consequently, their responses could introduce errors or irrelevant information (referred to as “noise”) that do not accurately reflect the constructs the survey is intended to measure [48]. Taken together, these factors may lead to a misrepresentation of the true nature of the phenomena under investigation. Such misrepresentations could have occurred during the course of this research, potentially affecting the validity and reliability of the findings. Future studies should consider incorporating observational or experimental methods to validate self-reported behaviors.

Further, although the research model attained a relatively high R-squared value of 0.64, suggesting that it accounts for a substantial portion of the variation in security behavior, there may be additional unexplored psychological factors influencing security behavior that were not captured in the model. Future research could extend explanatory power by incorporating constructs such as optimism bias and subjective norms. In the discussion, several speculative explanations for the findings were also raised, including the potential influence of email use frequency, optimism bias, and subjective norms on perceived susceptibility, the impact of perceived severity on perceived importance, the role of digital literacy in perceived barriers, and the effect of cybersecurity awareness on self-efficacy. While these interpretations offer plausible insights, they remain speculative and require empirical testing to confirm their validity.

An additional point of interest is that 42.4% of respondents reported using email “rarely” while 23.2% reported having been victims of phishing. Although this was beyond the scope of the current analysis, future studies could explore this relationship from different perspectives to determine whether lower email use frequency reduces familiarity with phishing cues, thereby paradoxically increasing susceptibility.

Future research should also consider validating the effect of prior cyber trauma or attack. The construct of prior cyber trauma or attack would capture the experiences of individuals who have been victims of cyberattacks or cyber-related traumatic events and the psychological and emotional impact of those experiences. Such a construct could help explain how past victimization shapes perceptions of susceptibility, severity, benefits, barriers, and self-efficacy in the cybersecurity context. Integrating prior trauma into the Health Belief Model could therefore enrich understanding of how lived experiences influence protective security behaviors.

Although CFA and SEM provided a robust test of the hypothesized relationships and confirmed the validity of the measurement model, these methods are inherently constrained by their reliance on theory-driven, linear specifications at the group level. They cannot capture nonlinear dynamics, unanticipated interactions, or provide explanations at the individual level. Future research could therefore extend this work by incorporating explainable artificial intelligence (XAI) approaches that are capable of uncovering hidden patterns, revealing subgroup heterogeneity, and generating interpretable explanations for individual variations in security behavior. The data set established in this study, already validated through CFA and SEM, offers a valuable foundation for such extensions. Subject to appropriate anonymization and ethical safeguards, the data can be made available for research purposes upon reasonable request.

Finally, while niche studies targeting specific demographics are encouraged and offer valuable insights, they also limit generalizability. Findings from this study, for example, reflect the behaviors and perceptions of a single demographic group of university students in Indonesia and cannot be assumed to apply broadly. Future research should extend to other likely susceptible groups to ensure a more comprehensive understanding of the phenomenon across different contexts and populations.

Conclusions

The study explored several hypotheses regarding the relationship between various constructs derived from HBM and security behavior in the context of email phishing attacks among university stu-

dents. Notably, the hypotheses can be categorized into those accepted and those rejected based on the empirical findings. The results supported hypotheses regarding perceived severity (H2), perceived importance (H3), cues to action (H5), and self-efficacy (H6), revealing statistically significant positive correlations with security behavior. This suggests that individuals who perceive phishing attacks as severe, recognize the importance of proactive measures, respond to action cues, and have confidence in their abilities are more likely to engage in security behaviors. Perceived severity, self-efficacy, and cues to action showed moderate significance, while perceived importance exhibited strong significance. However, hypotheses concerning perceived susceptibility (H1) and perceived barriers (H4) were not supported, indicating no significant correlation with security behavior.

Theoretically, the findings pose a challenge to the conventional focus of the HBM on perceived susceptibility and barriers as primary behavior predictors. This underscores the necessity to assess the relevance of HBM constructs in security behavior, indicating that contextual factors may influence their significance. Additionally, the robustness of the HBM-based model, evidenced by its high R-squared value of 0.64, suggests that around 64% of the variance in security behavior can be accounted for by the predictor variables. This implies that variables like perceived susceptibility, severity, importance, barriers, cues to action, and self-efficacy collectively offer significant explanatory power in understanding security behavior in phishing attack scenarios. Hence, the HBM framework effectively captures the complexity of human decision-making in cybersecurity contexts. Moreover, considering the emphasis on human behavior as the weakest link in cybersecurity, this research insights aid in addressing this weakness by understanding their behavior. Thus, the research answers the call for more human-centered solutions and behavioral cybersecurity research where behavioral theories, in this case, HBM, are explored to understand human behavior in the information security context.

As for practical implications, the study insights can inform the design of cybersecurity initiatives, whether conventional or novel like gamification, that leverage individuals’ perceptions to either reinforce or discourage specific behaviors. By incorporating factors such as perceived severity, importance, self-efficacy, and cues to action, cybersecurity initiatives can effectively target the psychological determinants which influence security behaviors, informing the development of much more effective security awareness initiatives for the study population.

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Author contributions

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Appendix A

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Table A1. Student Cybersecurity Awareness Survey (SCAS) constructs and corresponding survey items.

Construct	Item	Source
Perceived Susceptibility (SUS)	SUS1: There is a good possibility that my personal information (login credentials, bank account details, etc.) gets stolen and misused due to an email phishing attack.	[61,67]
	SUS2: I feel that my chances of receiving an email phishing attack are high.	[67]
	SUS3: I will likely be a victim of an email phishing attack.	[67]
	SUS4: I worry a lot about becoming victimized in an email phishing attack.	[67]
	SUS5: It is likely that my computer becomes infected by a virus because of an email phishing attack.	[61,67]
Perceived Severity (SEV)	SEV1: Losing my personal information (login credentials, bank account details, etc.) due to an email phishing attack is a serious problem for me.	[59,67]
	SEV2: Having my computer infected by a virus because of an email phishing attack is a serious problem for me.	[59,67]
	SEV3: If I become a victim of an email phishing attack, my daily work could be negatively affected.	[59,67]
	SEV4: The thought of becoming a victim of an email phishing attack scares me.	[67]
	SEV5: Phishing attacks by email are harmful.	[67]
Perceived Importance (IMP)	IMP1: Taking action to prevent email phishing attacks is important for me.	[66,67]
	IMP1: Exercising care when reading emails with links is important for me.	[67]
	IMP2: Educating myself about email phishing attacks is important for me.	[67]
	IMP3: Using tools to prevent email phishing attacks is important for me.	[67]
	IMP4: Staying alert for email phishing is important for me.	[67]
Perceived Barriers (BAR)	BAR1: Exercising care when reading emails with links would require starting a new habit, which is difficult.	[59,67]
	BAR2: Educating myself about phishing takes too much time.	[67]
	BAR3: I don't know how to find and get the right tools or software to prevent email phishing attacks.	[67]
	BAR4: don't know what to look for to detect phishing emails.	[67]
	BAR5: I am afraid I would not be able to detect phishing emails.	[67]
Self-efficacy (SEF)	SEF1: I am confident of recognizing a phishing email.	[59,67]
	SEF2: I can recognize a phishing email even if there was no one around to help me.	[59,67]
	SEF3: I can recognize a malicious URL from a legitimate URL.	[67]
	SEF4: I am sure of the steps to follow to recognize a phishing email.	[67]
Cues to Action (CUE)	CUE1: If I saw a news report or awareness poster about phishing, I would be more conscious of potentially falling victim to an email phishing attack.	[61,67]
	CUE2: If I noticed a suspicious transaction on my bank account, I would be concerned about being a victim of an email phishing attack.	[67]
	CUE3: If I noticed suspicious log-in attempts on my social media account(s), I would be concerned about being a victim of an email phishing attack.	[67]
	CUE4: If my family were to tell me of a recent experience with an email phishing attack, I would be more conscious of potentially falling victim to an email phishing attack.	[61,67]
	CUE5: If my fellow students were to tell me of a recent experience with an email phishing attack, I would be more conscious of potentially falling victim to an email phishing attack.	[61,67]
Security Behaviour (BEH)	BEH1: I exercise caution when I receive an email with a link.	[59,67]
	BEH2: Before clicking on a link in an email, I will first check if the sender and subject of the email make sense.	[59,67]
	BEH3: Before clicking on a link in an email, I will first check if the URL is legitimate.	[59,67]
	BEH4: I do not click on a link in an email if the content of the email looks suspicious.	[59,67]
	BEH5: I report phishing emails to help people avoid becoming victims.	[67]

Appendix B

Table B1. Descriptive analyses results.

Gender	Frequency	%	Cumulative %
Male	244	45.6	45.6
Female	291	54.4	100.0
Total	535	100.0	
Study level	Frequency	%	Cumulative %
Bachelor	514	96.1	96.1
Masters	21	3.9	100.0
Total	535	100.0	
Nature of study program	Frequency	%	Cumulative %
IT-related program	205	38.3	38.3
Non-IT related program	330	61.7	100.0
Total	535	100.0	
Received cybersecurity awareness from university	Frequency	%	Cumulative %
No	267	49.9	49.9
Yes	268	50.1	100.0
Total	535	100.0	
Email use frequency	Frequency	%	Cumulative %
Several times a week	181	33.8	33.8
Daily	78	14.6	48.4
Rarely	227	42.4	90.8
Once a week	44	8.2	99.1
Never	5	.9	100.0
Total	535	100.0	
Knowledge of phishing attacks	Frequency	%	Cumulative %
No	73	13.6	13.6
Yes	459	86.3	100.0
Total	535	100.0	
Victim of phishing attack	Frequency	%	Cumulative %
No	411	76.8	76.8
Yes	124	23.2	100.0
Total	535	100.0	

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